History of Construction Cultures – Mascarenhas-Mateus & Paula Pires (eds)
© 2021 Copyright the Author(s), ISBN 978-1-032-00203-3.
Open Access: www.taylorfrancis.com, CC BY-NC-ND 4.0 license

Wooden churches, managers and Fulbright scholars: Glued laminated timber in 1950's Norway

M. Rusak
Arkitektur- og designhøgskolen i Oslo, Oslo, Norway

ABSTRACT: The paper explores the introduction of glued laminated timber technology to the Norwegian construction market in the late 1950s. The paper concentrates on the activities of Norwegian construction conglomerate Moelven Brug, which in the 1950s brought back the production of laminated timber in Norway after 40 years of absence in the country. Building on the work of Norwegian glulam pioneer Guttorm Brekke (1885–1980) and on state-of-the-art academic research and industry know-how from the United States of America, Sweden, and Belgium, in the post-war period Moelven Brug quickly established itself as the market leader for laminated timber in Norway. By mapping these scholarly and professional exchanges, the paper traces the shifting ontological understanding of timber in Norway from a "natural" to a "scientific" material and its transformation into a complex assemblage of different substances subject to control and standardization through a series of variables and parametric values.

1 INTRODUCTION

1958 was the year of glulam in Europe. At the World Fair in Brussels in April, an exceptional number of pavilions featured structures in glued laminated timber (Devos & Floré 2009). The same year, Johannes Mageli, general manager of the Norwegian construction company Moelven Brug visited another "eye-opening" exhibition of building materials in Stockholm. There, upon examining products from Töreboda Träkonstruktioner, he realized that technology of lamination fitted like a "hand in a glove" with both the products and facilities of Moelven Brug (Mageli 1996). Laminated load-bearing elements complimented the already-developed prefabricated products and could be used in schools, gyms, storage and factory buildings. In fact, this Swedish factory whose production inspired Mageli was set up by a Norwegian civil engineer Guttorm Brekke some 40 years prior. By August 1958, a study group financed by Moelven visited several factories in Holland and Belgium to examine the production of engineered timber and to evaluate whether it would be possible to transfer this technology to Norway (Skogstad 2000). The group included Moelven's general manager, the head of local forest cooperative and two representatives from the Norwegian Institute of Wood Technology (NTI). These four individuals represented the main stakeholders in the laminated timber industry: material producers, research institutes, and foreign experts. The agency of all these actors was instrumental in re-establishing glued laminated timber technology in Norway.

This paper examines the network of academic, professional and industrial actors fostered by the Fulbright and European Recovery Plan programs between the United States of America and Norway in the 1950s. In particular, the paper considers how the introduction of this new material in Norway transformed local timber traditions, prompting a reconceptualization of timber from being a "natural" Norwegian material to a "scientific" one controlled through a series of variables. In addition, the paper considers the ways in which the import of the process of lamination was adapted by Moelven Brug to local conditions of labor, technical knowledge and supply of raw materials. The company became the main actor that advanced engineered timber on the Norwegian market, fostered the growth of new regulation bodies and crafted a unique marketing narrative bridging tradition and modernity that persists today.

2 THE OLD-NEW MATERIAL

Although gluing wood has been known for as long as people have used timber, the first contemporary use of lamination was recorded in 1893 in Basel, Switzerland (Johansen & Engan 2001). In 1904, a Weimar firm Otto Hetzer AG patented a glued compound beam and in 1907 a glued laminated arch (Hetzer 1907). Gluing together several layers of wood increased its structural resilience, forming a single roof member and accommodating spans up to 10 meters long. Technology of lamination overcame shortcomings of traditional construction methods, "showing that tradition is not always a blessing" (Brekke 1945). Over time, engineered timber proved more economic and efficient in terms of construction time in comparison with steel:

DOI 10.1201/9781003173434-208

it could accommodate large spans, while its structures were light and quick to assemble.

Otto Hetzer held licenses for laminated construction in Germany, Switzerland, and Denmark, but in the post-World War One period he nearly entirely abandoned the technology of gluing in favor of prefabricated connector-built trusses (Brekke 1945). This shift could be largely attributed to the fact that laminated technology made extensive use of a casein-based glue – a derivative of milk products – and with the post-war rationing of milk and complicated economic situation in Germany it was impossible to obtain the necessary amounts of raw ingredients (Brekke 1945; Rhude 1998). Since Norway and Sweden remained neutral throughout World War One, this situation allowed industrial producers to continue experimenting with lamination processes. In the absence of their own steel facilities and with shortages of rolled steel available on the market, glued laminated timber represented an attractive structural alternative.

In Norway, glued laminated timber was introduced by an engineer Guttorm Brekke who studied in Germany with Otto Hetzer and in 1916 at a cost of 60,000 NOK bought a patent for the use of laminated timber constructions in Norway, Sweden, "North Russia south to Vilna" and the United States of America (Brekke 1945). Overall, Brekke's patent rights were exercised only in Norway and Sweden between the years 1917 and 1925. The purchase of the license included a two-month course at the Hetzer AG factory in Weimar and a set of detailed technical instructions for all production aspects, besides the recipe of the glue (Brekke 1945). In 1918, Brekke established a laminated timber company Trækonstruktion A/S in Mysen, outside Oslo, with all equipment and facilities modelled after the Hetzer factory (Brekke 1945). For example, an assembly hall of 60 by 100 feet had plain wooden floors so that curves of glued laminated arches could be drawn on a one-to-one scale. In addition, the factory included large storage and wood conditioning facilities, a dry kiln, and a small steel workshop, along with hoists, monorails and mechanical equipment necessary for assembly. The technology transfer from Germany was so precise that Brekke obtained all equipment – from a special combined mixer and a grinder for manufacture of casein glue to "excellent" screw clamps – directly from Weimar (Brekke 1945).

Glulam elements produced by the company found broad application: from a pavilion for a Nordic-American exhibition in Tivoli in 1919, famous experimental "American" houses at the Norwegian Technical University in 1922 to railway stations in Oslo, Trondheim and Lillestrøm and a Norwegian pavilion at the world exhibition in Rio-de-Janeiro in 1922–3 (Bugge 1922). Among the most popular products were a truss design system adapted for churches, auditoriums and gyms and two- and three-hinged girders. In Norway, these systems found broad application in industrial facilities: for example, a storage building at Meraker Brug, a chemical factory at Sarpsborg and a zinc-rolling mill at Sannesund. In 1919, Brekke opened



Figure 1. Fribärande Träkonstruktioner construction for Sundsvall station hall, Sweden.

another company Fribärande Träkonstruktioner AB in Töreboda, Sweden that was the first in Sweden to implement fan-driven kilns that allowed for faster drying. The company worked closely with the expanding national railway agency SJ and delivered train stations and platform buildings including large halls for Stockholm, Malmö and Gothenburg (Figure 1). In addition to railway buildings, the factory produced sports and tennis halls – including the King's tennis hall – theaters, factory and storage facilities with just over 300 structures built in the first couple of years (Brekke 1945).

However, casein glue was not entirely water-resistant and Brekke's glulam structures showed some signs of decay under prolonged exposure to damp and rain – an inherent quality of Scandinavian climate. Slow chemical hydrolysis during exposure to water, biological activity during too high humidity and temperature and mechanical stresses induced by shrinking and swelling caused deterioration of casein glue joints (Brekke 1945). The effectiveness of casein glue was hard to test or improve based on shared experiences, as most of the casein glue recipes remained a concealed business secret. In general, the glulam industry was at a disadvantage in Scandinavia, as the operating costs were high and the products "had not yet gained the confidence of the public" (Brekke 1945). With increased competition from concrete and steel producers that gradually returned to the market in the post-war period, many factories went out of business. With large operating debt and general liquidity problems, the first venture into laminated construction in Norway came to a halt. In 1925 it was decided to discontinue the production at Mysen, and when the factory burned down a year after, it was never rebuilt. Träkonstruktioner in Sweden with its established name and well-developed technology was sold, while Brekke decided to emigrate to the USA.

Although early history of the 20th century glued laminated constructions is well-studied in continental Europe, particularly in Belgium, Switzerland, and Germany, in Scandinavia this history still needs to be written (Devos & Floré 2009; Rinke 2015, 2019; Rinke

& Haddadi 2018, 2020). Virtually no works in Norwegian or Swedish historiography trace proliferation of this new building material. This research thus primarily draws on archival sources that have never been examined before, primarily Brekke's report (Brekke 1945). Guttorm Brekke emigrated to the USA in the late 1920s and worked for the National Bureau of Standards in Washington DC. In 1945, he was commissioned by the Norwegian Industrial Committee in New York to assess possibilities for re-introducing glued laminated timber to post-World War Two Norway (Amdam 2000). Comprising more than 700 pages featuring the early history of lamination, intricacies of Brekke's own venture and assessment of the most recent international research, this 1945 report is a unique first-hand historical account. Largely overlooked by architecture and construction historians, this report ought to provide the foundation in the historiography of engineered timber in Norway.

Guttorm Brekke was a key figure of a pioneering calibre, joining the present and the past: products from his Töreboda factory exhibited in Stockholm in 1958 inspired Mageli to try this technology at Moelven Brug. Brekke's report, in turn, contained valuable technical information, assessed the possibility for reintroducing this technology in Norway and was well known to technical professionals, advising Moelven on the intricacies of production.

3 AN AMERICAN REVIVAL

European in its origins, with advanced testing and research facilities in the USA, glued laminated timber was gradually transformed into a material perceived as "American" (Skjelmerud 1952). Imported to the USA with waves of inter-war European immigration, initially it was adapted to a limited extent before World War Two. Max S. Hanisch – a German émigré and another student of Otto Hetzer – was among the pioneers of glulam in the USA. In 1934, Hanisch founded Unit Structures Inc. and completed the first building from laminated wooden elements in the US – Peshtigo High School gymnasium (Fischetti & McNall 1995). Initially, innovative timber constructions were not approved by the Wisconsin Industrial Commission and Hanisch sought assistance from the USDA-Forest Products Laboratory (FPL) in Madison, Wisconsin. Although not familiar with the glulam elements before, FPL became the first institution to experiment with glued laminated timber in the USA. In 1934, a joint research program was started between the laminating plant of Unit Structures at the FPL (Rhude 1998).

While in Europe the main driver for lamination was shortage of other structural materials, in the United States the technology primarily took off as a way to utilize around 300 million acres of new forests. Research of the Forest Product Laboratory thus also aligned with the strategy of finding new markets for existing forest resources (FPL 1938). FPL studied mechanical properties of wood in tension, bending and compression, degrees of shear, toughness and rigidity (Brekke 1945). These stress tests provided accurate knowledge of timber performance, helped to avoid the "irrational" use of materials and defined the discrepancy between different wood species – a factor never accounted for before (FPL 1938). Most importantly, however, FPL enhanced and developed new resin glues that came from Europe and were used in the aircraft industry. New added catalysts made low-temperature setting resin glues possible to apply to thicker laminated elements, thus simplifying the lamination process and making structures more durable. The findings of the laboratory on new water-resistant glues were published "with typical American grandeur" and patented for the benefit of the people, in stark contrast to the "atmosphere of secrecy which has been customary in Europe" (Brekke 1945).

The laboratory experimented with new adhesives and technological details of the lamination process, recording results of these studies in exhausting tabulations of parameters and numerical values, graphs and force diagrams (Brekke 1945). Similarly to the way Michael Osman argues that concrete had been transformed from a material of "liquid stone" derived from natural components to a material of analytical control and managerial organization, it is possible to argue that laboratory experiments with new lamination processes transformed timber from a "natural" to a "scientific" material whose structural qualities and production process could be meticulously controlled and adjusted through a series of parametric variables (Osman 2012). Further, American research on synthetic resin adhesives, glue solvents and catalysts, fire retardants and chemical seasoning significantly improved the technological process of lamination and made engineered timber competitive with other post-war construction materials.

Knowledge exchange between Europe and the Unites States went both ways. As glued laminated timber was a novelty in the Unites States, American specialists visited European structures to study their structural qualities and performance. For example, in the summer of 1936, Mr T.R.C. Wilson, Senior Engineer of the FPL inspected around 50 glued laminated structures in Norway, Sweden, Denmark, Germany and Switzerland. In particular, Wilson visited one of the train platform cover structures built by Brekke in Trondheim in 1921, and later wrote that "[it] is in excellent condition as are structures in Sweden and Switzerland" (Brekke 1945). This served as proof that although European specialists were concerned with the waterproof qualities of the glue used for lamination, in fact these structures showed little signs of wear, "a proof which time only could provide" (Brekke 1945).

Thus, although European in its origins, the technology of lamination was significantly updated with state-of-the-art American research that challenged the role of timber as a "natural" material and enhanced it with non-traditional qualities, like resistance against fire, fungi and decay. While research on engineered

timber was transferred to the USA with waves of European immigration, the technology was developed and returned to the continent with enhanced qualities, most importantly, waterproof resin adhesives.

4 INDUSTRIAL-ACADEMIC RESEARCH

In the years following World War Two Norwegian businesses fostered close academic and industrial ties with the USA, particularly through projects financed by the Technical Assistance and Productivity program (also known as the Marshall Plan). From singular actors to entire educational institutions, the latest research in construction technology and particularly glued laminated timber was transferred back to Norway. Norsk Treteknisk Insititutt (NTI) – the Norwegian Institute of Wood Technology – was established in March 1949, largely following the example set by the Forest Products Laboratory.

To compensate for intellectual isolation during the Second World War the institute immediately ordered a substantial timber research library from the USA with generous financial support offered by Marshall Aid (Skogstad 2000). The institute's leader – engineer Halvor Skjelmerud – spent a year in the USA "educating himself about different aspects of wood technology and developing contacts," while NTI established a bilateral research collaboration with FPL (Skjelmerud 1952). Starting from 1955, many Fulbright scholars associated with NTI went to the USA, boosting the Norwegian-American knowledge exchange (Skogstad 2000). In 1957, FPL specialist Magnus L. Selbo – a key scholar in the field of lamination in the USA – came to NTI as part of a professional exchange program and authored a report on lamination of structural members together with Norwegian Ole Grønvold (FPL 1958). Engineer Hvamb went for a year's study in the US and came back with an impression that the laminated industry was the only one working at full capacity in the shrinking timber market, concluding that lamination harbored a lot of potential for Norway (Skjelmerud 1958). In turn, Brekke's report with its extensive technical research was well-known to the Institute's specialists.

In this way during the years after World War Two research on glued laminated timber made its way back to Norway both through the practice of individual actors – such as Brekke – and large scale international academic exchange, facilitated by Marshall aid. The Norwegian Institute of Wood Technology was the first body in the country to conduct comprehensive research on glulam, testing material properties of different timber species, water-resistant adhesives and drying technologies. Research findings were assembled in a vast technical library of reports, available for any private company to consult (Skogstad 2000). Managers of Moelven Brug – upon considering a venture into glued laminated timber – contacted NTI for consultation, and developed a close relationship with the Institute from the very beginning.

Halvor Skjelmerud – NTI's leader – was very enthusiastic about Moelven's venture and offered technical and information support. According to Skjelmerud, "it [was] not clear to [NTI] what the reservations about the introduction of [lamination technology] could be," as there were "no special Norwegian circumstances" that would hinder the development of this technology in Norway. However, "technical conservatism" of the Norwegian construction branch could be hard to overcome, warned Skjelmerud (Skjelmerud 1952). Two Institute representatives – Karl Mørkved and Eirik Raknes – were part of a Moelven study trip in 1958 that went abroad to study factories in Holland and Belgium that worked with laminated technology. Johannes Moe – a civil engineer from NTI who specialized in concrete constructions in the US – assisted Laminator with the research on the dimensioning for laminated constructions and, according to Mageli, provided "invaluable assistance in the problematic starting period" (Mageli 1996). The Institute's engineers, in fact, were instrumental in making Moelven laminated load-bearing structural members efficient and economic.

NTI was not the only research institution that Moelven developed close connections with. The company also sought advice from Hans Granum – a civil engineer, the head of the construction technology at the architecture department at NTH and the main specialist on modern timber structures in Norway. Granum suggested that laminated beams would be particularly suitable for typologies where the load-bearing constructions had to be visible and that aesthetics of laminated beams would be of a particular interest for architects (Granum 1958). Among the greatest advantages Granum identified "beautiful, concentrated form and significant height," concluding that laminated constructions would be particularly suitable in "more modern architecture where clear material effects and simple forms are sought after" (Granum 1958). In addition, he also pointed out that laminated beams could be used for agricultural barns with an overall market of about 1,5 million NOK – an important typology for the Norwegian economy that strove to modernize its agriculture. In this way upon seeking to implement the new technology of lamination, Moelven Brug fostered close industrial-academic ties with various institutions and research actors.

Besides seeking advice from academic researchers, Moelven managers were also interested in industrial experience with lamination: they studied technical processes at factories throughout Europe and the United States and fostered new business connections. However, not all experience could be applied to the Norwegian context: during visits to Nemaho facility in Holland, despite a seemingly amicable agreement with factory administration, Norwegian representatives only got a glimpse of raw material storage and some ready-made constructions without getting any specific information about the technology, equipment or production process (Mageli 1996). Swedish factories of Töreboda seemed old and technically outdated, as their equipment remained unchanged since the

times of Brekke (Mageli 1996). Moelven American partners – Rilco products, the largest glulam producer in the USA – were supposed to supply Norwegian partners with technical data and calculations but this ambitious venture proved to be of little use (Mageli 1959). With its costly machinery, the technological process abroad had little application to Norwegian realities. While cutting-edge equipment could have been bought in the United States, Moelven Brug was short on cash, could not afford expensive equipment and thus had to engage in-house mechanical engineers to develop necessary machinery, like a new hydraulic press (Mageli 1996).

In this way although the lamination technology was perceived as one imported from abroad and despite extensive efforts to gather international industrial experience, in reality, little technical knowledge was transferred directly. Thus, "foreign" technology had to be adapted to available material and financial resources, local labor market and the conditions of the Norwegian building industry.

5 LOCAL APPLICATION

As Moelven established a subsidiary company Laminator A/S that specialized in load-bearing structures built with glued laminated timber, the production was strongly dependent on the local supply of raw materials. In the late 1950s, Hedmark and Oppland – two regions where the company was located – had the lowest timber prices in the country and the best-quality pine forest, making it easy for the company to access raw timber from a well-developed network of local sawmills at a low price (Vevstad 1994). The price difference for raw materials between Moelven's native municipality and other regions was so significant that it offset additional transportation costs required to deliver ready-made products to Oslo and its vicinity, fostering a close connection between the industry and its locality (Mageli 1958).

To ensure uninterrupted access to high-quality timber, Laminator was established as a shared enterprise with Ringsaker Almenning – a local forest cooperative that supplied all raw materials and pine elements used for lamination and had modern drying facilities. The lamination process in turn had to be adjusted to the physical properties of the local spruce and pine found in Ringsaker forests (Bovim & Sund 1977). In addition, Moelven's new factory had a formative effect on the local social fabric as it provided jobs for workers left unemployed by the increasing mechanization of Norwegian forestry and agriculture. As the process of lamination required a particular skill-set, the company considered the uneducated population an advantage, as new employees could be taught necessary skills quicker (Lindberget 1963). In this way, the technology of lamination was not just imported, but also adapted to the availability of local materials and labor.

"Laminated wood constructions from Moelven used for barns as well as churches" – opened one of the



Figure 2. Moelven laminated timber advertisement.

articles on Laminator products from 1960 (Ringsaker Blad 1960). And indeed, they were: load-bearing glulam beams had broad applications in typical Norwegian architecture of the everyday. The company delivered cheap, efficient and reliable structures that fulfilled their purpose at minimal cost. Laminator produced elements for schools, barns, churches, sports halls, storage buildings, hangars and airports – any typology that required large spans (Figure 2).

Curiously, churches became one of the more peculiar buildings designed by the company. The first church in the country to be built with laminated timber was Jar Kirke in Bærum in 1959. The company delivered 23 straight 17-meter-long beams for the church's load-bearing roof constructions (Hamar Arveiderblad 1959). The following year, a chapel was built in Østfold, another church in Søre Ål in 1963, chapels in Ramsund and Borge in 1964 and a Storhamar church in 1971. By the 1970s, the company developed a typological solution for a modern Norwegian church that relied on laminated load-bearing constructions and walls built from large prefabricated elements (Sveen 1975). Moelven's prefabricated solution offered a possibility to obtain a representative structure at a low price.

Throughout the 1960s, Laminator was actively marketing the use of laminated timber in religious buildings, referring to the updated tradition of stave churches: "modern church architecture uses wood, which is both strong and decorative," while Laminator beams could be used for all structures "whenever the construction is visible" (Ringsaker Blad 1960). Exposed glulam beams offered a pragmatic but representative solution for an important element

of Norwegian everyday life and drew parallels with the centuries-long tradition of wooden churches. Thus, Laminator elements provided an opportunity for new architectural expression, merging a traditional material with new technology, marketed particularly to architects and design professionals. For example, Norwegian famous architect Sverre Fehn used Moelven laminated timber beams in his famous Cathedral Museum of Hedmark (Domkirkeodden, Hamar 1988). The company extensively used the fact that Moelven beams made their way into signature architecture in their advertisements throughout the 1970s (Dalseg-Ervaco 1976).

In addition, "foreign" technology was ready to be domesticated for the local needs of the pragmatic welfare state. Many of Laminator's clients were large state and municipal actors, as the company worked on extending its connections within the state apparatus. Laminated timber beams were particularly suitable for sports halls, and became popular when provision of sports infrastructure became one of the priorities of Norwegian education. Rational and pragmatic, built with unusual shapes and the exposed materiality of wood, these sports halls soon became local landmarks. The first gym in laminated timber was finished in 1962 at Persbråten school with arches that spanned 40 meters (Hamar Arbeiderblad 1962). Another significant commission was Askerhallen – by then, the largest covered ice-stadium in the country with a ground plate of 105 by 51 meters, the fourth indoor ice-hockey rink ever built in Norway and the first in the Oslo area. Designed by a Fredrikstad architect Aksel Fronth, all timber elements were delivered by Laminator.

As with churches, Laminator developed a standardized typological solution and "Moelven-halls" were delivered all over the country – "from Karasjok in the north to Egersund in the south". They were fast to erect and cheap in comparison with facilities built using traditional methods. Typology of sports halls – similar to churches – combined utility and representation, while the use of laminated timber offered to bridge structural engineering and architectural work. In this way, by 1967 laminated timber became one of the most sought-after construction materials, while laminated timber structures became an increasingly common sight within representative buildings of everyday Norwegian life (Ringsaker Blad 1967). This development was likely a result of Moelven's successful marketing narrative that balanced claims of rationality of construction and expressive architectonic effects, efficiency and aesthetics, technology and tradition (Figure 3).

The story of updated tradition crafted for architects and design professionals culminated in the competition for the Lillehammer Olympics in 1994. After Moelven Brug won the commission and built three large Olympic stadiums in laminated wood, Moelven glulam products became intertwined with ideas of "contemporary Norwegian building style" that merged the wooden tradition and new technology. It was precisely this narrative recognized abroad that was key in Moelven receiving a commission for the new international Gardermoen airport in Oslo.

6 CONCLUSION

Re-introduction of glued laminated timber to the Norwegian construction market in the post-1945 period involved a wide range of actors: from industrial producers and academic researchers to foreign experts and marketing managers. A traditional Norwegian resource, timber was transformed from a "natural" material into a "scientific" one enhanced with new structural and material properties and strengthened by synthetic adhesives. Moelven Brug was a key actor in bringing the technology of glulam back to Norway, actively building academic and industrial connections locally and internationally. Although perceived as borrowed from abroad, international experience with lamination had to be adapted to Norwegian material and labor conditions. Incorporated into significant representative typologies, such as churches, glued laminated timber proved to be ambiguous in terms of the cultural interpretations attributed to it. Moelven managed to harness those interpretations to introduce structural timber into a wide variety of representative building structures across the country, making laminated timber synonymous with a "new Norwegian building style" (Karlsen 1999). An elusive bridge between tradition and modernity offered to a professional audience was a curious marketing by-product that fed into a myth of an industrialized timber tradition that flourishes until today.

As engineered timber was making its way to the Norwegian construction market, new regulation bodies overseeing the material quality and implementation had to be established. In 1962, Moelven Brug initiated a foundation of Lamineringsutvalg – a self-assessment organization of laminate producers, public officials and researchers with a goal to assure compliance within the new industry. This oversight body became international in 1965 as Limtre Norden gathered similar actors from Sweden, Denmark and Finland. By the late 1970s as Moelven grew into a large group of companies, Moelven Limtre – a new name for Laminator A/S – bought most competitors not just within Norway, but also Scandinavia (Mageli 1996). Among them was Fribärande Träkonstruktioner AB – a former Brekke company in Sweden and one whose products inspired Mageli to venture into laminated technology. In this way the historical loop was closed: today, Moelven Limtre positions itself as a direct descendant of Brekke's venture and retrospectively claims the design of all train station projects undertaken by Fribärande Träkonstruktioner (Brandt 2019). Moelven Brug was not just the main actor who brought laminated timber back to Norway, but also one who eventually completely shaped the market and defined implementation of this technology in the north.

This paper out to map the main actors involved in the establishment of the glued laminated industry



Figure 3. Moelven laminated timber advertisement.

in Norway, although its limited scope does not allow giving justice to all figures involved. While the 20th century history of lamination technology in Scandinavia remains to be written, this paper makes the first step toward uncovering this fascinating international exchange of ideas.

REFERENCES

Amdam, R.P. 2000. Industrikomiteen i New York 1943–1945: ein kanal for kunnskapsoverføring frå USA til Norge. *Historisk tidsskrift* 79(1): 3–21.

Bovim, N.I. & Sund, H. 1977. *Limtreboken.* Moelv: Moelven Limtre.

Brandt, K. 2019. Active centenarian that changed the industry. *Wood Magazine* 3.

Brekke, G.N. 1945. *Glued laminated timber for the building industry, Report no.2.* Industrikomitéen i New York, S-2079/E/Eb/L0034, Box 0010, Folder 0001-C-111r. Riksarkivet, Oslo, Norway.

Bugge, A. 1922. *Forsøkshuser: opført ved Norges tekniske høiskole, Trondhjem.* Trondheim: F. Bruns bokhandels forlag.

Dalseg-Ervaco. 1976. *Byggekunst* 1.

Devos, R. & Floré, F. Modern Wood. 2009. De Coene at Expo 58. *Construction History* 24: 103–20.

Fischetti, C. & McNall, A. 1995. Glued Laminated Timber. In Jester. T.C (ed.) *Twentieth-century Building Materials: History and Conservation*. University of Michigan: McGraw-Hill.

Forest Products Laboratory (FPL). 1938. *The Forest Products Laboratory. A brief account of its work and aims.* U.S. Department of Agriculture.

Forest Products Laboratory (FPL). 1958. *Forest Products Laboratory Research Program.* Madison: FPL.

Hetzer, O. 1907. *UK Patent No 20684*.

Håkenåsen, J. 1982. *Ringsakeralmenningenes historie / utgitt av Ringsakeralmenningene*. Ringsaker: Ringsakeralmenningene.

Johansen, K. & Engan, F. 2001. *Liming av tre*. Nesbru: Vett & viten AS.

Karlsen, A. 1999. *Institusjonelle perspektiver på næringsomstilling*. Trondheim: Norbok.

Lindberget, E. 1963. Poenget ikke å starte stort, men at bedriften blir lønnsom. *Aftenposten*, 8 October, 9.

Mageli, J. 1958a. *PM til styret i A/S Moelven Brug*. Moelven Industrier archive, ARK-287-02, Box E, Folder Ea. State Archive in Hamar, Hamar, Norway.

Mageli, J. 1958b. *PM vedr. Opprettelse av fabrikk i Moelv for limte trekonstruksjoner*. Moelven Industrier archive, ARK-287-02, Box E, Folder Ea. State Archive in Hamar, Hamar, Norway.

Mageli, J. 1959. *PM til styret i A/S Moelven Brug*. Moelven Industrier archive, ARK-287-02, Box E, Folder Ea. State Archive in Hamar, Hamar, Norway.

Mageli, J. 1996. *A/S Laminator. Etablering, vekst og utvikling.* Moelven Industrier archive, ARK-287-02, Box E, Folder Ea. State Archive in Hamar, Hamar, Norway.

Osman, M. 2012. The managerial aesthetics of concrete. *Perspecta* 45: 67–76.

Rhude, A. 1998. Structural Glued Laminated Timber: History and Early Development in the United States. *APT Bulletin* 29 (1): 11–17.

Rinke, M. 2015. Terner & Chopard and the new timber. Early Technological Development and Application of Laminated Timber in Switzerland. In *Proceedings of the Fifth International Conference on Construction History*. Chicago: Construction History Society of America.

Rinke, M. 2019. Mechanization and early hybrid material use in glulam construction – The tram depot in Basel from 1916. In *Proceeding of the Sixth Conference of the Construction History Society, Cambridge: Construction History Society*. Cambridge: University Press.

Rinke, M. & Haddadi, R. 2018a. The riding arena in St. Moritz and the locomotive depot in Bern - a comparative study of early glulam construction in Switzerland. In *Proceeding of the Sixth Conference of the Construction History Society, Cambridge: Construction History Society*. Cambridge: University Press.

Rinke, M. & Haddadi, R. 2018b. Transforming the traditional timber roof — the sport hall in Birsfelden as an early glulam application in Switzerland. In *Iron, Steel and Buildings: the Proceedings of the Seventh Conference of the Construction History Society*. Cambridge: University Press.

Skjelmerud, H. 1952. *Rapport fra besøk ved tretekniske forskningsinstitusjoner og trelastog treindustrier i det vestlige USA og Canada*. Oslo: Norsk Treteknisk Institutt.

Skjelmerud, H. 1958. *Correspondence with Moelven Brug, Oslo, 22 August 1958*. Moelven Industrier archive, ARK-287-02, Box E, Folder Ea. State Archive in Hamar, Hamar, Norway.

Skogstad, P. 2000. *Kunnskap for fremtiden: Norsk treteknisk institutt 1949–1999.* Oslo: Norsk Treteknisk Institutt.

Sveen, Willy. 1975. Storhamar Kirke. *Byggekunst* 57(6): 154–157.

Untitled. 1959. A-S Laminator i Moelv en suksess allerede før produksjonen er kommet ordentlig i gang. *Hamar Arbeiderblad*, 16 October, 3.
Untitled. 1960. Den første Laminator-låven reiser seg i Ringsaker. Ringsaker Blad, 23 August, 1,4.
Untitled. 1962. 40 meters takbuer fra Moelv. *Hamar Arbeiderblad*, 29 November, 1.
Untitled. 1967. For låge jernbaneunderganger flaskehals for limtre fra Moelv. *Ringsaker Blad*, 28 December, 2.
Verdensutstillingen. 1924. *Beretning om Norges deltagelse i Verdensutstillingen i Rio de Janeiro, 1922–1923*. Kristiania: Norbok.
Vevstad, Andreas. 1994. *Saa skal alminding være: Norsk almenningsforbund 1919–1994*. Elverum: Norsk almenningsforbund.